

WARTH, Austria

WMO: 113080

Lat: 47.256N

Lon: 10.186E

Elev: 1470

StdP: 84.86

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.7	(c) -13.3	(d) -18.4	(e) 0.9	(f) -13.9	(g) -16.6	(h) 1.0	(i) -11.3	(j) 6.5	(k) 1.8	(l) 5.7	(m) 1.2	(n) 1.0	(o) 80	(p) 0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.6	(c) 24.0	(d) 15.2	(e) 22.2	(f) 14.4	(g) 20.5	(h) 13.5	(i) 16.4	(j) 22.2	(k) 15.3	(l) 20.6	(m) 14.3	(n) 19.2	(o) 2.0	(p) 350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.2	12.1	18.9	13.1	11.3	17.3	12.3	10.6	16.4	51.7	22.3	48.1	20.9	45.2	19.1	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.9	4.2	3.4	-18.6	26.5	2.9	1.7	-20.7	27.8	-22.4	28.8	-24.0	29.8	-26.1	31.0
			-18.9	17.5	2.8	1.3	-20.9	18.4	-22.5	19.2	-24.0	19.9	-26.1	20.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-3.2	-2.4	0.2	3.4	7.8	11.6	13.2	13.1	9.5	7.0	1.5	-1.7		
	DBStd	7.27	4.55	5.03	4.55	4.17	3.80	4.32	3.57	3.39	3.65	4.16	5.15	4.85		
	HDD10.0	2185	410	348	304	199	89	32	13	11	53	107	256	363		
	HDD18.3	4863	668	581	562	448	327	206	163	164	266	351	505	622		
	CDD10.0	371	0	0	0	2	20	80	111	107	37	14	1	0		
	CDD18.3	8	0	0	0	0	0	3	3	2	0	0	0	0		
	CDH23.3	76	0	0	0	0	0	24	28	24	0	0	0	0		
	CDH26.7	4	0	0	0	0	0	3	1	1	0	0	0	0		
(14) Wind	WSAvg	1.4	1.5	1.5	1.5	1.4	1.4	1.3	1.3	1.2	1.2	1.2	1.3	1.3		
(15) Precipitation	PrecAvg	1844	120	120	139	115	170	201	218	211	155	120	134	140		
	PrecMax	2415	297	367	266	239	412	337	388	406	307	260	339	345		
	PrecMin	1408	3	24	45	19	45	84	87	66	55	13	1	12		
	PrecStd	206	79	73	67	55	74	62	78	86	57	57	72	75		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	9.8	12.5	16.6	21.4	26.1	26.1	25.7	21.5	19.2	14.2	9.9		
		MCWB	3.1	2.9	5.0	8.8	12.2	15.4	15.7	15.9	13.4	10.7	7.6	3.8		
	2%	DB	6.3	7.8	10.5	14.5	18.9	23.3	23.8	23.7	19.5	16.9	12.2	7.7		
		MCWB	1.9	2.2	4.2	7.4	10.9	14.6	15.2	15.6	12.6	9.5	6.2	2.6		
	5%	DB	4.6	5.9	8.7	12.5	16.7	21.4	21.8	21.5	17.7	15.1	10.4	5.8		
		MCWB	1.1	1.1	3.3	6.2	9.8	14.0	14.4	14.4	11.9	8.9	5.3	1.7		
	10%	DB	3.0	4.1	6.8	10.4	14.6	19.3	19.7	19.6	15.9	13.2	8.3	4.1		
		MCWB	0.2	0.5	2.5	5.1	8.8	13.1	13.7	13.6	10.9	8.2	4.3	1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	3.9	5.8	9.7	13.5	19.0	17.4	17.6	15.0	12.1	8.4	4.6		
		MCDB	7.0	7.7	10.9	15.5	19.6	22.1	23.0	23.4	19.3	16.4	13.3	8.3		
	2%	WB	2.7	2.7	4.7	8.1	11.7	16.1	16.0	16.2	13.7	10.6	6.8	3.5		
		MCDB	5.2	6.2	9.1	13.4	16.9	22.3	22.1	22.0	17.7	14.8	10.9	6.3		
	5%	WB	1.6	1.8	3.8	6.7	10.5	14.4	15.0	15.1	12.6	9.7	5.6	2.5		
		MCDB	4.0	5.1	7.9	11.4	15.2	20.1	20.3	20.2	16.5	13.6	9.5	5.0		
	10%	WB	0.5	0.8	2.9	5.5	9.4	13.3	14.1	14.1	11.5	8.8	4.6	1.5		
		MCDB	2.7	3.8	6.1	9.7	13.7	18.4	18.7	18.6	14.9	12.4	8.0	3.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	7.9	8.0	8.4	8.6	8.6	8.6	8.4	8.8	7.2	6.6		
		MCDBR	8.2	9.5	10.7	11.3	12.2	12.3	12.3	12.4	11.8	11.8	9.6	8.3		
		MCWBR	5.4	6.0	6.0	6.0	5.9	5.6	5.5	5.8	6.3	6.1	5.6	5.2		
		MCWBR	7.3	8.4	9.5	10.3	10.4	10.9	10.8	11.3	10.5	10.1	8.9	7.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.235	0.256	0.290	0.324	0.337	0.344	0.344	0.338	0.312	0.289	0.262	0.234		
		taud	2.414	2.367	2.357	2.377	2.412	2.425	2.417	2.454	2.513	2.561	2.543	2.454		
		Ebn at Noon	883	928	938	931	926	918	913	906	904	881	846	844		
		Edh at Noon	65	86	102	112	112	111	111	102	88	71	58	56		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.35	2.27	3.40	4.33	4.85	5.51	5.45	4.79	3.61	2.42	1.40	1.12		
		RadStd	0.16	0.28	0.41	0.60	0.53	0.52	0.50	0.50	0.40	0.28	0.20	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (103 neighbors)		N/A	N/A	N/A	+0.64	+0.45	+0.40	N/A	N/A	+53	+13

Nomenclature: See separate page